E5CC

OMRON

Digital Controller

EN INSTRUCTION MANUAL

Thank you for purchasing the OMRON E5CC Digital Controller. This manual describes the functions, performance, and application methods needed for optimum use of the product. Please observe the following items when using the product.

- This product is designed for use by qualified personnel with a knowledge of electrical systems.
- Before using the product, thoroughly read and understand this manual to ensure correct use.
- Keep this manual in a safe location so that it is available for reference whenever required.

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Refer to the *E5□C Digital Controllers User's Manual* (Cat. No. H174) for detailed application procedures.

Safety Precautions

●Key to Warning Symbols

⚠ CAUTION	Indicates a potentially hazardous situation which, if not avoided, is likely to result in minor or moderate injury or property damage. Read this manual carefully before using the product.

CC1 1654637-7E

●Warning Symbols

⚠ CAUTION

Minor injury due to electric shock may occasionally occur. Do not touch the terminals while power is being supplied.	⚡
Electric shock, fire, or malfunction may occasionally occur. Do not allow metal objects, conductors, cuttings from installation work, or moisture to enter the Digital Controller, the Setup Tool ports, or between the pins on the connectors on the Setup Tool cable.	⊘
Do not use the product where subject to flammable or explosive gas. Otherwise, minor injury from explosion may occasionally occur.	⊘
Never disassemble, modify, or repair the product or touch any of the internal parts. Minor electric shock, fire, or malfunction may occasionally occur.	⊘
CAUTION - Risk of Fire and Electric Shock a) This is the product UL listed as Open Type Process Control Equipment. It must be mounted in an enclosure that does not allow fire to escape externally. b) More than one disconnect switch may be required to de-energize the equipment before servicing. c) Signal inputs are SELV, limited energy. d) Caution: To reduce the risk of fire or electric shock, do not interconnect the outputs of different Class 2 circuits.	⚠
If the output relays are used past their life expectancy, contact fusing or burning may occasionally occur. Always consider the application conditions and use the output relays within their rated load and electrical life expectancy. The life expectancy of output relays varies considerably with the output load and switching conditions.	⚠
Loose screws may occasionally result in fire. Tighten the terminal screws to the specified torque of 0.43 to 0.58 N•m.	❗
Set the parameters of the product so that they are suitable for the system being controlled. If they are not suitable, unexpected operation may occasionally result in property damage or accidents.	❗
A malfunction in the Digital Controller may occasionally make control operations impossible or prevent alarm outputs, resulting in property damage. To maintain safety in the event of malfunction of the Digital Controller, take appropriate safety measures, such as installing a monitoring device on a separate line.	❗

Suitability for Use

Omron Companies shall not be responsible for conformity with any standards, codes or regulations which apply to the combination of the Product in the Buyer's application or use of the Product.

At Buyer's request, Omron will provide applicable third party certification documents identifying ratings and limitations of use which apply to the Product. This information by itself is not sufficient for a complete determination of the suitability of the Product in combination with the end product, machine, system, or other application or use. Buyer shall be solely responsible for determining appropriateness of the particular product with respect to Buyer's application, product or system. Buyer shall take application responsibility in all cases.

NEVER USE THE PRODUCT FOR AN APPLICATION INVOLVING SERIOUS RISK TO LIFE OR PROPERTY WITHOUT ENSURING THAT THE SYSTEM AS A WHOLE HAS BEEN DESIGNED TO ADDRESS THE RISKS, AND THAT THE OMRON PRODUCT(S) IS PROPERLY RATED AND INSTALLED FOR THE INTENDED USE WITHIN THE OVERALL EQUIPMENT OR SYSTEM.

Precautions for Safe Use

Be sure to observe the following precautions to prevent operation failure, malfunction, or adverse affects on the performance and functions of the product. Not doing so may occasionally result in unexpected events.
Use the product within specifications.

(1) The product is designed for indoor use only. Do not use the product outdoors. Do not use or store the product in any of the following locations.
- •Places directly subject to heat radiated from heating equipment.
- •Places subject to splashing liquid or oil atmosphere.
- •Places subject to direct sunlight.
- •Places subject to dust or corrosive gas (in particular, sulfide gas and ammonia gas).
- •Places subject to intense temperature change.
- •Places subject to icing and condensation.
- •Places subject to vibration and large shocks.

(2) Use/store within the rated temperature and humidity ranges. Provide forced-cooling if required.
(3) To allow heat to escape, do not block the area around the product. Do not block the ventilation holes on the product.
(4) Be sure to wire properly with correct polarity of terminals.
(5) Use the specified size of crimped terminals (M3, width 5.8 mm or less) for wiring. To connect bare wires to the terminal block, use copper braided or solid wires with a gage of AWG24 to AWG18 (equal to cross- sectional area of 0.205 to 0.8231 mm²). (The stripping length is 6 to 8 mm.) Up to two wires of same size and type, or two crimped terminals can be inserted into a single terminal.
(6) Do not wire the terminals which are not used.
(7) Allow as much space as possible between the controller and devices that generate a powerful high- frequency or surge. Separate the high-voltage or large-current power lines from other lines, and avoid parallel or common wiring with the power lines when you are wiring to the terminals.
(8) Use this product within the rated load and power supply.
(9) Make sure that the rated voltage is attained within two seconds of turning ON the power using a switch or relay contact. If the voltage is applied gradually, the power may not be reset or output malfunctions may occur.
(10) Make sure that the Digital Controller has 30 minutes or more to warm up after turning ON the power before starting actual control operations to ensure the correct temperature display.
(11) When executing self-tuning, turn the load and the unit ON simultaneously, or turn the load ON before you turn the controller ON.
(12) A switch or circuit breaker should be provided close to this unit. The switch or circuit breaker should be within easy reach of the operator, and must be marked as a disconnecting means for this unit.
(13) Wipe off any dirt from the Digital Controller with a soft dry cloth. Never use thinners, benzine, alcohol, or any cleaners that contain these or other organic solvents. Deformation or discoloration may occur.
(14) Design system (control panel, etc) considering the 2 second of delay that the controller's output to be set after power ON.
(15) The output will turn OFF when you move to the Initial Setting Level. Take this into consideration when performing control.
(16) The number of non-volatile memory write operations is limited. Therefore, use RAM write mode when frequently overwriting data during communications or other operations.
(17) When disassembling the Temperature Controller for disposal, use suitable tools.
(18) Do not exceed the communications distance that is given in the specifications and use the specified communications cable. Refer to the *E5□C Digital Controllers User's Manual* (Cat. No.H174) for the communications distance and cable specifications.
(19) Do not turn the power supply to the Digital Controller ON or OFF while the USB-Serial Conversion Cable is connected. The Digital Controller may malfunction.
(20) The maximum terminal temperature is 75°C.

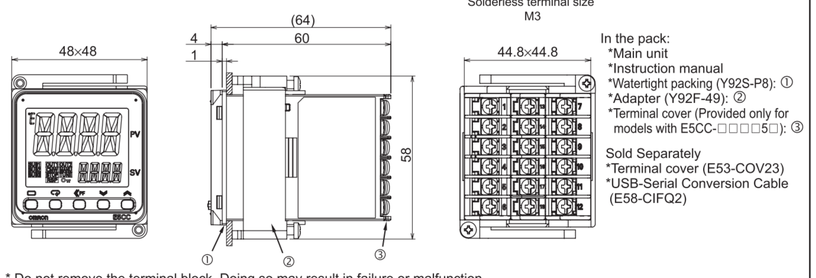
●Specifications

Power supply voltage	100 to 240 VAC, 50/60 Hz or 24 VAC, 50/60 Hz / 24 VDC
Operating voltage range	85 to 110% of the rated voltage
Power consumption Option 000: All other specifications:	 5.2 VA max. (100 to 240 VAC) 3.1 VA max. (24 VAC)/1.6 W max. (24 VDC) 6.5 VA max. (100 to 240 VAC) 4.1 VA max. (24 VAC)/2.3 W max. (24 VDC)
Indication accuracy (Ambient temperature: 23°C)	Thermocouple: (±0.3 % of indication value or ±1°C, whichever is greater) ±1 digit max. Platinum resistance thermometer: (±0.2 % of indication value or ±0.8°C, whichever is greater) ±1 digit max. Analog input: ±0.2 % FS ±1 digit max.
Event input Contact input No-contact input	 Output current: approx. 7 mA per contact. ON:1 kΩ max., OFF: 100 kΩ min. ON: residual voltage 1.5 V max., OFF: leakage current 0.1 mA max.
Remote SP input	4 to 20 mA DC or 0 to 20 mA DC 0 to 5 V DC or 1 to 5 V DC or 0 to 10 V DC
Control output 1	Relay output :SPST-NO 250VAC, 3A(resistive load) Electrical life of relay: 100,000 operations Voltage output (for driving SSR): 12 VDC ±20%, 21 mA Current output: 4 to 20 mA DC, 0 to 20mA DC Load: 500 Ω max.
Control output 2	Voltage output (for driving SSR): 12 VDC, 21 mA
Control method	ON/OFF or 2-PID control
Auxiliary outputs	Relay outputs:SPST-NO, 250 VAC, two outputs: 3 A (resistive load), three outputs: 2 A (resistive load), Electrical life of relay: 100,000 operations
Transfer output	4 to 20 mA DC with load of 500 Ω max. 1 to 5 VDC with load of 1 kΩ min.
Ambient temperature	-10 to 55°C (Avoid freezing or condensation)
Ambient humidity	25% to 85%
Storage temperature	-25 to 65°C (Avoid freezing or condensation)
Altitude	Max. 2,000m
Recommended fuse	T2A, 250 VAC, time-lag, low-breaking capacity
Weight	Approx. 120 g (Digital Controller only)
Degree of protection	Front panel: IP66 Rear case: IP20, Terminal section: IP00
Installation environment	Installation category II, pollution degree 2 (as per IEC61010-1)
Memory protection	Non-volatile memory (Number of write operations: 1,000,000)
Temporary overvoltage	Short term: 1200 V+ (power supply voltage) Long term: 250 V+ (power supply voltage)

■Wiring

●Dimensions

Dimensions (mm)



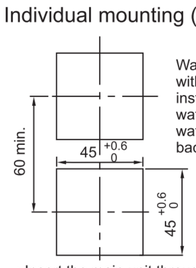
In the pack:
*Main unit
*Instruction manual
*Watertight packing (Y92S-P8): ①
*Adapter (Y92F-49): ②
*Terminal cover (Provided only for models with E5CC-□□□□5□): ③

Sold Separately
*Terminal cover (E53-COV23)
*USB-Serial Conversion Cable (E58-CIFQ2)

* Do not remove the terminal block. Doing so may result in failure or malfunction.
* A Setup Tool port is provided on the upper of the product. Use this port to connect a personal computer to the product when using the Setup Tool. E58-CIFQ2 USB-Serial Conversion Cable is required to connect the personal computer to the product. (Do not use the product with the USB-Serial Conversion Cable left permanently connected.) Refer to the instruction manual provided with the USB-Serial Conversion Cable for details on connection methods.

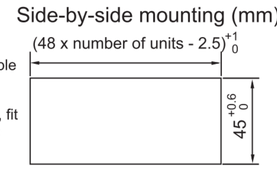
●Installation

Individual mounting (mm)



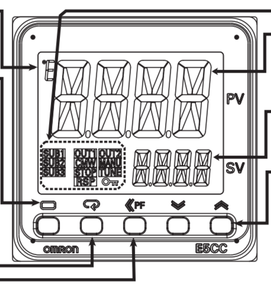
Waterproofing is impossible with side-by-side installation. When waterproofing is required, fit watertight packing on the backside of front panel.

Side-by-side mounting (mm)



- Insert the main unit through the mounting hole in the panel (1 to 5 mm thickness). Insert the mounting brackets (supplied) into the fixing slots located on the top and bottom of the rear case.
- Tighten the two mounting screws on the top and bottom of the adapter to keep them balanced, and finally tighten them to a torque of between 0.29 and 0.39 N·m.
- When more than one machine is installed, make sure that the ambient temperature does not exceed the specified limit.

●Names of Parts on Front Panel

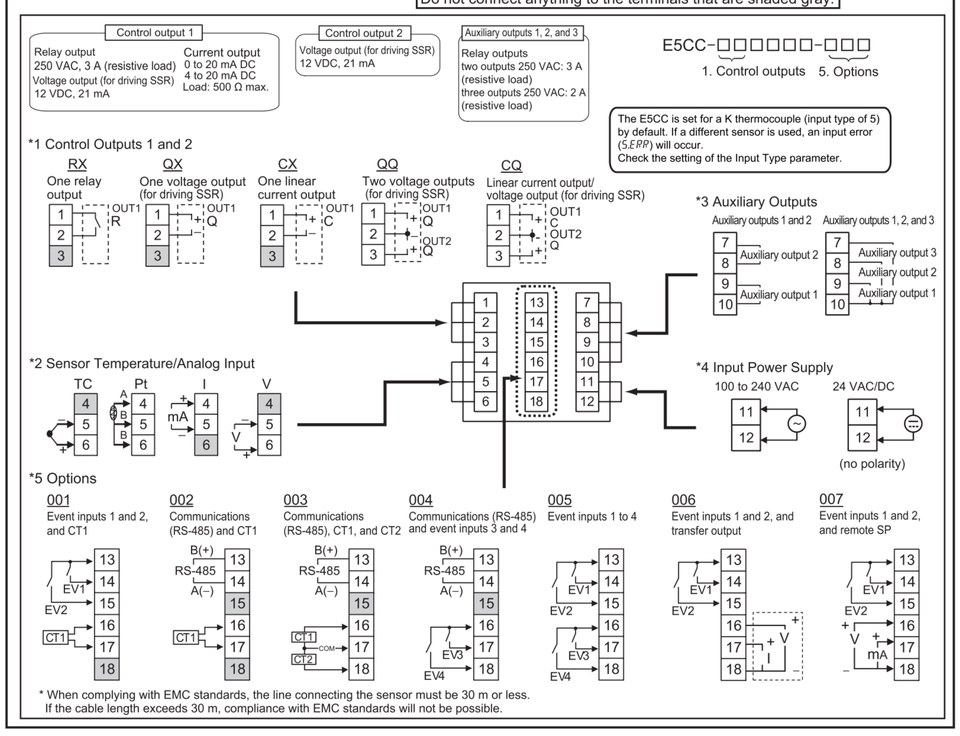


- °C / °F : temperature unit
  The temperature unit is displayed when the displayed value is a temperature. Either °C or °F is displayed according to the set value of the temperature unit.
- Level key
  Use this key to change levels:
- Mode key
  Press this key to change the contents of the display. Press this button for 1 s or longer for reverse scroll.
- Press the Mode key and the Up key together for at least 3 seconds to switch to protect level.
- Shift key (PF key)
  The default PF Setting parameter is for shifting the digit. This is a function key. When it is pressed, the function set for the PF Setting parameter will operate.
- No.1 display
  Process value or set data type
- No.2 display
  Set point, set data read-out value or changed input value
- Up and Down keys
  Each press of Up key increments or advances the values displayed on the No.2 display.
  Each press of Down key decrements or returns the values displayed on the No.2 display.

Operation indicators
- SUB1: Auxiliary output 1 indicator
- SUB2: Auxiliary output 2 indicator
- SUB3: Auxiliary output 3 indicator
- OUT1: Control output 1 indicator
- OUT2: Control output 2 indicator
- RSP: Remote SP indicator
  Lit when the assigned function is ON.
- TUNE:
  Flashing during self-tuning.
  Lit during auto-tuning.
- STOP: Control stopped indicator
  Lit when "Run/Stop" is stopped during operation. During control stop, functions other than control output are valid.
- CMW: Communications writing enabled/disabled indicator
  Lit when communications writing is enabled and not lit when it is disabled.
- Protection indicator
  Lit when Setting Change Protect is ON (disables the Up and Down Keys).
- MANU: Manual output indicator
  Lit when the Auto/Manual Mode is set to Manual Mode.

●Connections (The applicability of the electric terminals varies with the type of machine.)

Do not connect anything to the terminals that are shaded gray.



* When complying with EMC standards, the line connecting the sensor must be 30 m or less. If the cable length exceeds 30 m, compliance with EMC standards will not be possible.

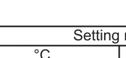
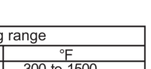
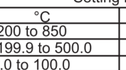
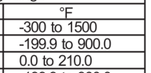
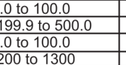
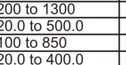
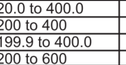
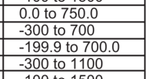
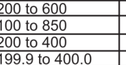
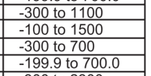
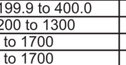
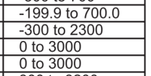
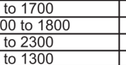
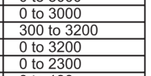
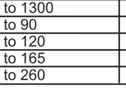
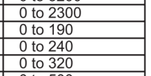
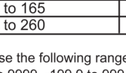
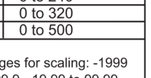
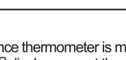
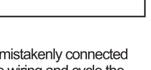
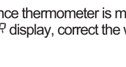
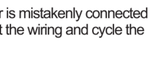
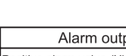
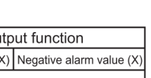
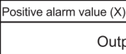
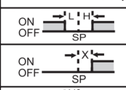
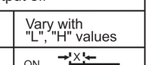
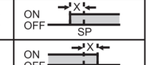
■Operation Menu

●Input Type

	Input type	Input	Setting	Setting range °C	Setting range °F
Temperature inputs	Platinum resistance thermometer	Pt100	0	-200 to 850	-300 to 1500
			1	-199.9 to 500.0	-199.9 to 900.0
		JPt100	2	0.0 to 100.0	0.0 to 210.0
			3	-199.9 to 500.0	-199.9 to 900.0
			4	0.0 to 100.0	0.0 to 210.0
	Thermocouple	K	5	-200 to 1300	-300 to 2300
			6	-20.0 to 500.0	0.0 to 900.0
		J	7	-100 to 850	-100 to 1500
			8	-20.0 to 400.0	0.0 to 750.0
		T	9	-200 to 400	-300 to 700
			10	-199.9 to 400.0	-199.9 to 700.0
		E	11	-200 to 600	-300 to 1100
		L	12	-100 to 850	-100 to 1500
		U	13	-200 to 400	-300 to 700
			14	-199.9 to 400.0	-199.9 to 700.0
		N	15	-200 to 1300	-300 to 2300
		R	16	0 to 1700	0 to 3000
		S	17	0 to 1700	0 to 3000
		B	18	100 to 1800	300 to 3200
		W	19	0 to 2300	0 to 3200
		PL II	20	0 to 1300	0 to 2300
	Infrared Thermosensor ES1B	10 to 70°C	21	0 to 90	0 to 190
		60 to 120°C	22	0 to 120	0 to 240
		115 to 165°C	23	0 to 165	0 to 320
		140 to 260°C	24	0 to 260	0 to 500
Analog input type	Current input	4 to 20mA	25	Use the following ranges for scaling: -1999 to 9999, -199.9 to 999.9, -19.99 to 99.99, -1.999 to 9.999	
		0 to 20mA	26		
	Voltage input	1 to 5V	27		
		0 to 5V	28		
		0 to 10V	29		

*The default is"5".
*5.ERR will be displayed when a platinum resistance thermometer is mistakenly connected while input type is not set for it. To clear the 5.ERR display, correct the wiring and cycle the power supply.

●Alarms

	Setting	Alarm type	Alarm output function: Positive alarm value (X)	Alarm output function: Negative alarm value (X)
	0	No alarm function	Output off	
*1	1	Deviation upper/lower limit		Vary with "L","H" values
	2	Deviation upper limit		
	3	Deviation lower limit		
*1	4	Deviation upper/lower range		Vary with "L","H" values
*1	5	Deviation upper/lower limit standby sequence ON		Vary with "L","H" values
	6	Deviation upper limit standby sequence ON		
	7	Deviation lower limit standby sequence ON		
	8	Absolute value upper limit		
	9	Absolute value lower limit		
	10	Absolute value upper limit standby sequence ON		
	11	Absolute value lower limit standby sequence ON		
	12	LBA (only for alarm 1)		
	13	PV Change Rate Alarm		
	14	SP absolute value upper limit		
	15	SP absolute value lower limit		
	16	MV absolute value upper limit		
	17	MV absolute value lower limit		
	18	RSP absolute value upper limit		
	19	RSP absolute value lower limit		

*1: Upper and lower limits can be set for parameters 1, 4 and 5 to provide for different types of alarm. These are indicated by the letter "L" and "H".
- The default alarm type is "2"

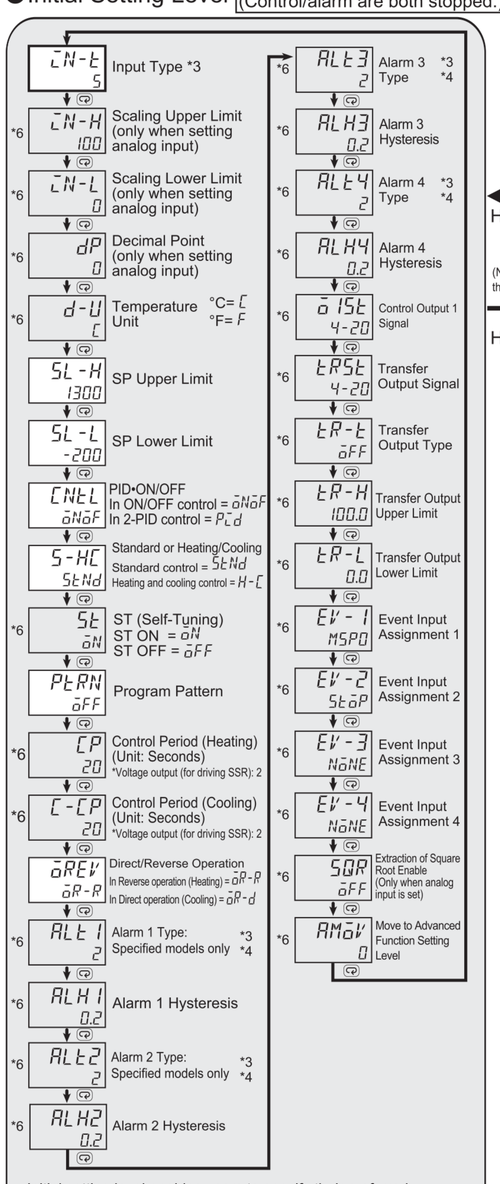
⚠ Conformance to EN/IEC Standards

This is a class A product.
In residential areas it may cause radio interference, in which case the user may be required to take adequate measures to reduce interference.

A 급 기기 (업무용 방송통신기자재)
이 기기는 업무용(A 급) 전자파적합기기로서 판매자또는 사용자는 이 점을 주의하시기 바라며, 가정외의지역에서 사용하는 것을 목적으로 합니다.

●Initial Setting Level

Operation stopped. (Control/alarm are both stopped.) *5



Initial setting level enables users to specify their preferred operating conditions (input type, alarm type, control method, etc.)

*3: Refer to the adjoining tables for details of input types and alarm types.
*4: Applicable only to models with alarm functions.
*5: Operation is stopped when moved to the initial setting level. (control/alarm are both stopped.)
*6: The grayed-out setting items are not displayed for some models and some settings of other setting items.
*7: Applicable only to models with heater burnout functions.
*8: The four numeric digits of the product code are displayed in the No. 2 display. The setting cannot be changed and there is nothing that you need to set.

⚠ Conformance to Safety Standard

Reinforced insulation is provided between input power supply, relay outputs, and between other terminals.

Do not allow temporary overvoltages on the primary circuit to exceed the following values.
Check the power supply voltage to the Digital Controller.
Short-term overvoltage: 1,200 V + (Power supply voltage)
Long-term overvoltage: 250 V + (Power supply voltage)

Always externally connect the recommended fuse that is specified in the Instruction Manual before you use the Digital Controller.

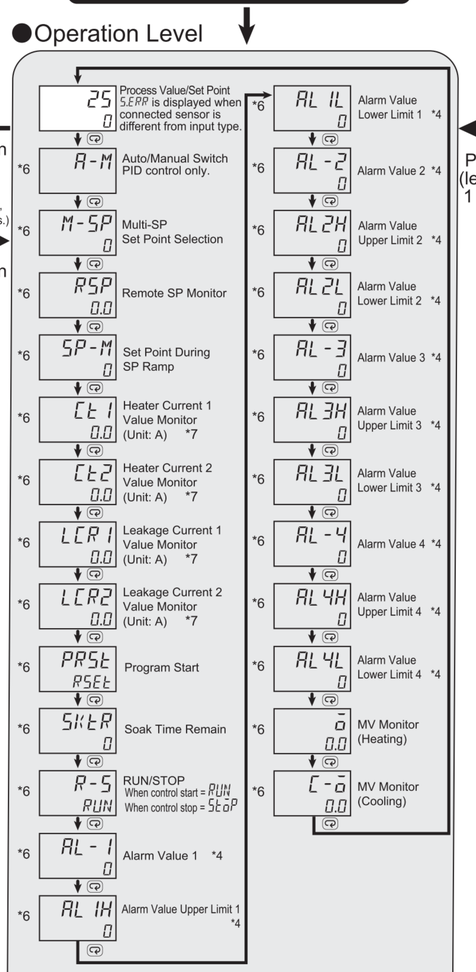
Analog Input
- If you input an analog voltage or current, set the Input Type parameter to the correct input type.
- Do not use the Digital Controller to measure a circuit with Measurement Category II, III, or IV.
- Do not use the Digital Controller to measure an energized circuit to which a voltage that exceeds 30 Vms or 60 VDC is applied.

The protection provided by the Digital Controller may be impaired if the Digital Controller is used in a manner that is not specified by the manufacturer.

Check the wiring before turning ON the power supply.

POWER ON

●Operation Level



Operation level should normally be used during operations.

Hold Mode down for at least 3 seconds (No.1 display flashes, then the control stops.)

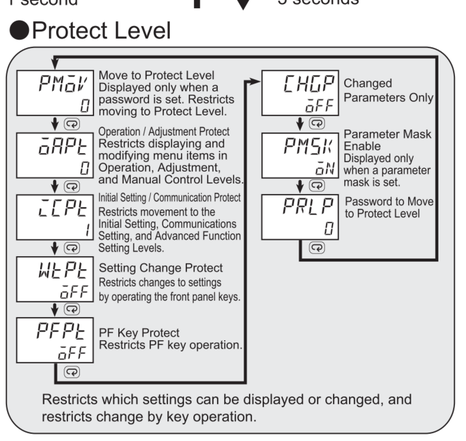
Hold Mode down for at least 1 second

Press Mode (less than 1 second)

Hold Mode and Up keys down for at least 1 second

Hold Mode and Up keys down for at least 3 seconds

●Protect Level



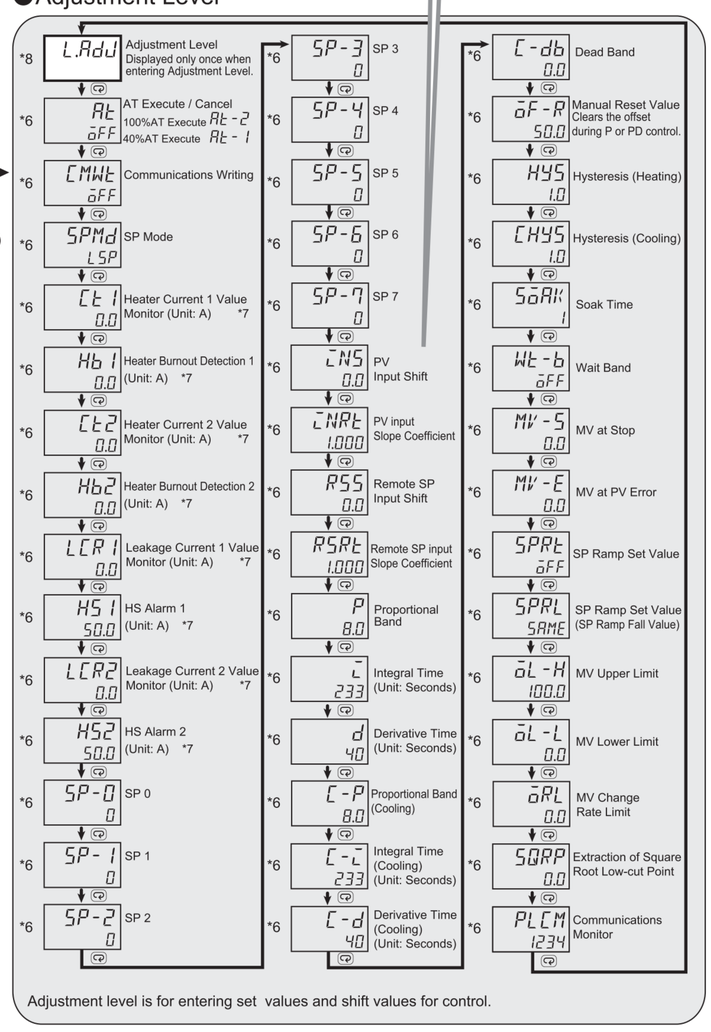
Restricts which settings can be displayed or changed, and restricts change by key operation.

■Other functions

Refer to the *E5□C Digital Controllers User's Manual* (Cat. No. H174) for information on the Advanced Function Setting Level, Manual Control Level, and other functions.
Refer to the *E5□C Digital Controllers Communications Manual* (Cat. No. H175) for information on communications.

Only the value set to the 2NS: Temperature Input Shift parameter is applied to the entire temperature input range. When the process value is 200°C, the process value is treated as 201.2°C after input shift if the input shift value is set to 1.2°C. The process value is treated as 198.8°C after input shift if the input shift value is set to -1.2°C.

●Adjustment Level



Adjustment level is for entering set values and shift values for control.

●Error Display (troubleshooting)

When an error has occurred, the No.1 display shows the error code. Take necessary measure according to the error code, referring the table below.

No.1 display	Meaning	Action	Status at error: Control output	Status at error: Alarm
5.ERR (S. Err)	Input error *2	Check the setting of the Input Type parameter, check the input wiring, and check for broken or shorts in the temperature sensor.	OFF	Operates as above the upper limit.
E333 (E333)	A/D converter error *2	After the correction of A/D converter error, turn the power OFF then back ON again. If the display remains the same, the controller must be repaired. If the display is restored to normal, then a probable cause can be external noise affecting the control system. Check for external noise.	OFF	OFF
E111 (E111)	Memory error	Turn the power OFF then back ON again. If the display remains the same, the controller must be repaired. If the display is restored to normal, then a probable cause can be external noise affecting the control system. Check for external noise.	OFF	OFF

If the input value exceeds the display limit (-1999 to 9999), though it is within the control range, cccc will be displayed under -1999 and ---- above 9999. Under these conditions, control output and alarm output will operate normally.
Refer to the *E5□C Digital Controllers User's Manual* (Cat. No. H174) for the controllable ranges.

*2: Error shown only for "Process value / Set point". Not shown for other status.

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